Efficient Computation of Mathematical Shape Descriptors for 3D Image Analysis

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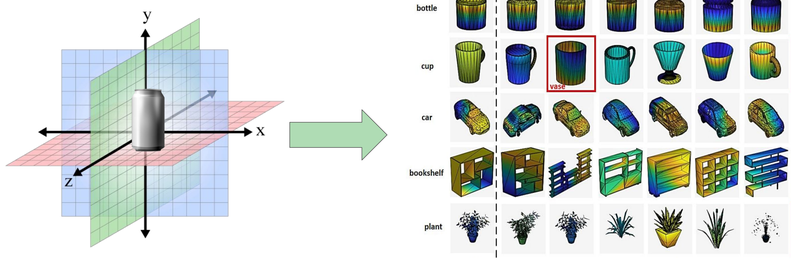
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Abstract

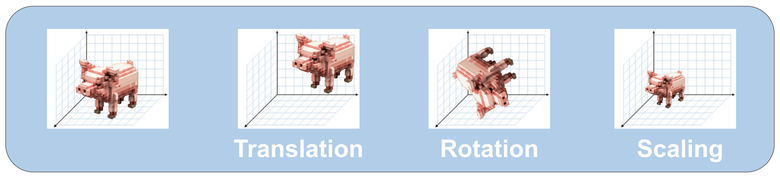
The ability to utilize three-dimensional (3D) image data, valuable for many disciplines and procedures such as medical cell imaging and biomolecular modeling, provides indispensable spatial information for technological and experiential developments. However, these tools and processes are computationally expensive and inefficient due to problems like scaling and rotation of such figures. In order to further develop these models, strong mathematical and computational methods must be advanced to effectively and accurately represent, compare, and segment this 3D data. Moments, generally, are quantitative measures that describe an image by using a set of mathematical functions. Due to their compact mathematical representation of images, certain variations are helpful to extract information about objects' features.

This project integrates fractional-order moment-based descriptors into the representation and comparison of 3D structural data of everyday objects. By mathematically manipulating and developing such tools, algorithms coded via Python convert images represented by 8 million voxels to just 121 unique numeric descriptors. Such 3D images, members of predetermined categories by shape, represent query objects that are unknown and in need of classification. The method's implementation includes database searching, various recurrence relations, reshaping of multi-dimensional arrays, and external storage of necessary functions, all after having verified the novel moments' orthogonality and precision. The program was refined for optimal time and storage efficiency. Subsequent calculations using a *k*-Nearest Neighbors (KNN) algorithm disclosed the method's accuracy for classifying objects depending on various parameters in the program. Upon completion, this effort will result in a widely applicable framework within which structural image data can be effectively retrieved, manipulated, mapped, and transferred to classification procedures.

Objective



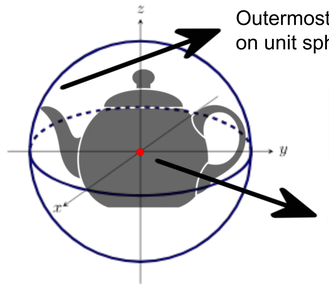
The long-term objective of this work is to develop a system that can classify unknown 3D objects within a large data set of known, labeled objects. Alternate methods are challenged by the variance caused by scaling, translation, and rotation of such objects, as exemplified below. Further, representation by voxelization is numerically cumbersome. Employing Zernike moments evades these setbacks, allowing an object made of 8 million voxels to be represented by just 121 numeric descriptors, making our method more compact, applicable, and efficient.



Methods



Preclassfied 3D images were acquired[1] and preprocessed by first being converted from 3D triangular mesh to cubic voxelization under distinct Cartesian functions. The objects were then made invariant to scaling and translation using geometric moments, formulated below, such that each fit precisely within the unit sphere with its center of mass at the origin.



$$M_{rst} = \iiint x^r y^s z^t f(x,y,z)\,dx\,dy\,dz$$

Objects' features were computed using fractional-order Zernike moments, a novel derivative of the tools first defined by Canterakis[2] with later developments by Novotni and Klein[3], which allow an image once described by 8 million voxels to be reduced to 1771 moments (below at left).

$$\Omega_{nl}^{m} = \frac{3}{4\pi} \sum_{r+s+t\le n} \overline{\chi_{nlm}^{rst}}\, M_{rst}$$

$$F_{nl} = \|\Omega_{nl}\|_2 = \sqrt{\sum_{m=-l}^{l} |\Omega_{nl}^{m}|^2}$$

By grouping these moments into vectors using the procedure shown above at right, the object's representation is compressed further into 121 unique numeric descriptors with rotational invariance.

$$Z_{nl}^{m,\alpha}(r,\theta,\phi) = P_{nl}^{\alpha}(r)\, e_l^m(\vec{\mathbf{x}})$$

Achieved by modifying the radial functions in the traditional polynomials, our new formulation– *fractional-order* Zernike moments– incorporates an additional parameter, $\alpha \in \mathbb{R}^+$. When $\alpha = 1$, the polynomials are simplified back to their original form.

Formulas for computations were developed and translated to Python code, exemplified at right, which is comprised of various recurrence relations designed for maximal efficiency of time and storage. Other elements of the algorithm include variable initialization, external storage of some coefficients, and reshaping of multi-dimensional arrays.

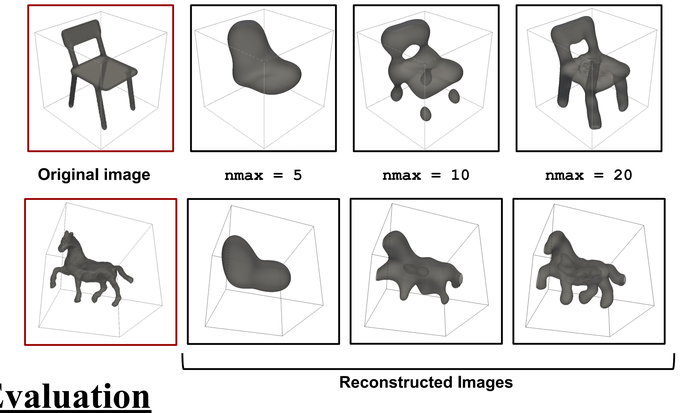
$$\chi_{nlm}^{rst} = c_l^m 2^{-m} \sum_{\nu=0}^{k} q_{kl}^{\nu} \sum_{\alpha=0}^{\nu} \binom{\nu}{\alpha} \sum_{\beta=0}^{\nu-\alpha} \binom{\nu-\alpha}{\beta} \sum_{u=0}^{m} \binom{m}{u} i^u (-1)^{m-u} \cdot \sum_{\mu=0}^{\lfloor \frac{l-m}{2} \rfloor} (-1)^{\mu} 2^{-2\mu} \binom{l}{\mu} \binom{l-\mu}{m+\mu} \sum_{\tau=0}^{\mu} \binom{\mu}{\tau},$$

```
# calculating chi
for J in range(K):
    n = indices[J][0]
    l = indices[J][1]
    m = indices[J][2]
    k = (n-l)/2
    k = int(k)

    if m >= 0:
      c = (sqrt((2*l + 1))*sqrt(factorial(l+m))
      *sqrt(factorial(l-m)))/factorial(l)
      coeff1 = c*2.0**(-m)
      for nu in range(0,k+1):
        q = ((-1)**k/2**(2*k))*sqrt((2*l+4*k+3)/3)
        *comb(2*k,k)*(-1)**(nu)*
         (comb(k,nu)*(comb(2*(k+l+nu)+1,2*k))
         /(comb(k+l+nu,k)))
        coeff2 = coeff1*q
        for alpha in range(0,nu+1):
          comb1 = comb(nu,alpha)
          coeff3 = coeff2*comb1
          for beta in range(0,(nu-alpha)+1):
            comb2 = comb(nu-alpha,beta)
            coeff4 = coeff3*comb2
            for u in range(0,m+1):
              comb3 = comb(m,u)*(1j**u)*((-1)**(m-u))
               #j used as imaginary unit
              coeff5= coeff4*comb3
              for mu in range(0,(floor((l-m)/2))+1):
                comb4= (-1)**(mu)*(2)**(-2*mu)
                 *comb(l,mu)*comb(l-mu,m+mu)
                 coeff6 = coeff5*comb4
                 t = 2*(nu - alpha - beta - mu) + l - m
                 for tau in range(0,mu+1):
                   comb5= comb(mu,tau)
                   coeff7=coeff6*comb5
                   r = 2*(tau + alpha) + u
                   s = 2*(mu - tau + beta) + m - u
```

Ultimately, each object, by its particular set of 121 numeric descriptors, can be compared element-wise for classification of images into respective categories by visual detail.

The model runs at a computation time comparable to that of traditional methods but with improved accuracy. Furthermore, our code's efficiency mechanisms have reduced run-time per image from 12 minutes to just over 7 seconds. Additionally, the predetermined maximum value of the parameter *n* has a direct effect on the algorithm's speed and precision, as seen below.



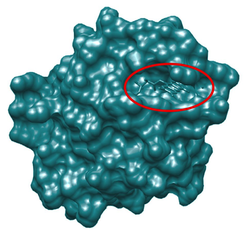
Evaluation

A *k*-Nearest Neighbors algorithm was applied to our method to evaluate and compare results according to the α variable. Given a query image, this strategy searches the dataset and ranks possible categories by a similarity score based on the Euclidean distance between the fractional Zernike descriptors of images. For this testing we used 200 randomizations, 10-fold cross validation, an 80/20 train/test split ratio, and scaling and stratification of data to reveal that $\alpha = 0.45$ yields the most optimal accuracy score overall when compared with $\alpha = 1$. We especially notice improvements in discriminative power for images with thin, elongated parts. However, we find that the best performing α varies from image to image, as this evaluation was conducted empirically for this particular dataset.

Number of Randomizations	200
Median Accuracy for $\alpha = 0.45$	86.2%
Median Accuracy for Traditional Descriptors	82.9%

Future Work and Applications

Areas of relevance include computational drug design and recognition of biological images, as well as facial recognition, climate research, and machine learning. For instance, these descriptors can represent cavities on the surface of proteins where ligands, like pharmaceuticals, can bind.



References

[1]X. Chen, A. Golovinskiy, and T. Funkhouser. A benchmark for 3D mesh segmentation. *ACM Transactions on Graphics* (Proc. SIGGRAPH), 28(3), 2009.

[2]N. Canterakis. 3D Zernike moments and Zernike affine invariants for 3D image analysis and recognition. *11th Scandinavian Conf. on Image Analysis*, 85–93, 1999.

[3]M. Novotni and R. Klein. 3D Zernike descriptors for content based shape retrieval. *Proceedings of the 8th ACM symposium on Solid modeling and Applications*, 216–225, 2003.

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